

# Likelihood Examples

## Today's agenda:

- Worked examples (chapter 8)

# Previously

We've talked about the different steps of the modelling process

Today, we'll walk through the whole process together

# First step, let's get some data

```
library(emdbook)
```

```
library(bbmle)
```

```
data("SeedPred")
```

What is in this dataset?

# First step, let's get some data

?SeedPred

Data from a seed predation experiment

Key variables are:

- Number taken
- Number Available
- Species
- Distance from forest edge

Our response variable is going to be the number taken.

What is our first step?

# Data cleaning/prep

Subset to only records with seeds available

```
SeedPred <- SeedPred[which(SeedPred$available > 0),]
```

Toss NAs

```
SeedPred <- na.omit(SeedPred)
```

Run these lines of code now.

What is next step?

# Choosing a distribution

What factors should we consider?

# Choosing a distribution

We can also visualize the distribution in different ways

```
hist()
```

But we can also use `barchart()` from the `lattice` package

This allows us to group the data in different ways using `table()`

# table()

We discussed table() a while ago

- Shows how common different categorical combinations are

Useful because it can use multiple levels of classification

```
table(SeedPred$taken)
```

```
table(SeedPred$taken, SeedPred$available)
```

Try these and see if the output makes sense. What is in each row/column?



## table() + barchart()

barchart() plays nicely with table(), creating nested barcharts

```
library(lattice)
```

```
barchart(table(SeedPred$taken), stack = FALSE)
```

```
barchart(table(SeedPred$taken, SeedPred$available), stack = FALSE)
```

Run these and examine the figures.

What's the most common number of seeds taken?

## table() + barchart()

Most of the time, 0 seeds are taken.

- This suggests we may need a special distribution

We can look at only those with more than zero taken.

```
nz <- SeedPred |> filter(taken > 0)
barchart(table(nz$taken, nz$available, nz$dist, nz$species),
          stack = FALSE)
```

Anything interesting about the distributions?

(Particularly at a distance of 25m, where 5 seeds are available)

# Picking a distribution

- We have lots of zeroes
- Some evidence of bimodality
- Values are between 0 and number available
- Binary and discrete (seeds are taken/not)

Possible options are binomial and beta-binomial

- BUT lots of zeroes suggests we may want a zero-inflated version
- We'll write functions (but note packages also contain pre-built versions)

# Zero-inflated functions

## Zero-inflated binomial

- Basically a `dbinom()` function modified to have a parameter for zero inflation (`zprob`)
- Where `zprob = 0`, it will just be a `dbinom()`

```
dzibinom <- function(x, prob, size, zprob, log = FALSE){  
  
  logv <- log(1 - zprob) + dbinom(x, prob = prob, size = size,  
                                  log = TRUE)  
  
  logv <- ifelse(x == 0, log(zprob + exp(logv)), logv)  
  
  if (log){logv}else{exp(logv)}  
  
}
```

Create this function now

# Zero-inflated functions

We'll also make a zero inflated beta-binomial (which would better handle bimodality)

```
dzibb <- function(x, size, prob, theta, zprob, log = FALSE) {  
  logv <- ifelse(x > size, NA, log(1 - zprob) +  
    dbetabinom(x, prob = prob,  
              size = size,  
              theta = theta,  
              log = TRUE))  
  logv <- ifelse(x == 0, log(zprob + exp(logv)), logv)  
  
  if(log){logv}else{exp(logv)}  
}
```

Create this function now

# Candidate distributions

We have 4 candidate distributions:

Binomial

Beta-binomial

Zero-inflated binomial

Zero-inflated beta-binomial

How do we figure out which one to use?

# Comparing distributions

```
SP.bb <- mle2(taken ~ dbetabinom( size = available,  
                                prob = prob,  
                                theta = theta),  
             start = list(prob = 0.5,  
                           theta = 1),  
             data = SeedPred)
```

```
SP.b <- mle2(taken ~ dbinom(size = available,  
                            prob = prob),  
            start = list(prob = 0.5),  
            data = SeedPred)
```

# Comparing distributions

For our zero-inflated models, we'll use a transformation

- `plogis()` keeps values between 0 and 1
- `qlogis()` will un-transform the values

**Try this out!** Pick some numbers and see what `plogis` gives you

```
tfed <- plogis(<any number>)  
untfed <- qlogis(tfed)
```



# Comparing distributions

```
SP.zibb <- mle2(taken ~ dzibb(size = available,  
                             prob = prob,  
                             theta = theta,  
                             zprob = plogis(logitzprob)),  
               start = list(prob = 0.5,  
                             theta = 1,  
                             logitzprob = 0),  
               data = SeedPred)
```

Run this and make sure it fits

# Comparing distributions

```
SP.zib <- mle2(taken ~ dzibinom(size = available,  
                                prob = p,  
                                zprob = plogis(logitzprob)),  
              start = list(p = 0.2,  
                           logitzprob = 0),  
              data = SeedPred)
```

Run this and make sure it fits

# Comparing distributions

Which model fits best? How do we tell?

# Comparing distributions

Which model fits best? How do we tell?

|         | dAIC   | df |
|---------|--------|----|
| SP.bb   | 0.0    | 2  |
| SP.zibb | 2.2    | 3  |
| SP.zib  | 419.5  | 2  |
| SP.b    | 3704.3 | 1  |

# Building our model

We've figured out which distribution fits best.

Now what?

# Seed predation and distance

To start, we'll ask whether distance from the forest changes predation

```
SP.bb.dist <- mle2(taken ~ dbetabinom(prob,  
                                     size = available,  
                                     theta),  
                  parameters = list(prob ~ dist - 1,  
                                     theta ~ dist - 1),  
                  start = as.list(coef(SP.bb)),  
                  data = SeedPred)
```

# Seed predation and distance

How do we examine our model output?

Which terms were significant?

Notice anything about the theta estimates?

# Seed predation and distance

Theta estimates are nearly identical

- Can we improve our model AIC by removing this parameter?
- How would we change out model to test this?



## Seed predation and distance

```
SP.bb.dist.theta <- mle2(taken ~ dbetabinom(prob,  
                                             size = available,  
                                             theta),  
                        parameters = list(theta ~ dist - 1),  
                        start = as.list(coef(SP.bb)),  
                        data = SeedPred)
```

Run this and compare it to SP.bb.dist

Which is the better model? Are they meaningfully different?

# Seed predation and distance model comparison

```
AICtab(SP.bb.dist,  
      SP.bb.dist.theta)  
  
anova(SP.bb.dist,  
      SP.bb.dist.prob)
```

What do the AIC and LRT tell us?

Does this make sense with the different theta estimates?

# Seed predation and distance model comparison

AIC says the models differ

ANOVA says the models aren't significantly different

The parameters are basically the same

- Probably no biologically meaningful differences in models

# Seed predation and distance model comparison

- Suggests that seeds closer to the forest are more to be taken
- No biologically meaningful difference in overdispersion ( $\theta$ )

# Seed predation and species

We also have information on species.

How would we test whether species differ in their responses?

# Seed predation and species

Modify our previous code:

```
SP.bb.sp <- mle2(taken ~ dbetabinom(prob = prob,  
                                     theta = theta,  
                                     size = available),  
                parameters = list(prob ~ species,  
                                   theta ~ species),  
                start = c(prob = -2,  
                          theta = -1),  
                data = SeedPred)
```

Run this. What happens?

# Seed predation and species

Modify with transformations (logis: -1 to 1; exp: 0 to +inf):

```
SP.bb.sp <- mle2(taken ~ dbetabinom(prob = plogis(lprob),  
                                     theta = exp(ltheta),  
                                     size = available),  
  parameters = list(lprob ~ species,  
                    ltheta ~ species),  
  start = c(lprob = -1,  
            ltheta = -1),  
  data = SeedPred)
```

Try running this version

# Seed predation and species

The model ran, now what should we do?



# Seed predation and species

Check the parameters using `summary(SP.bb.sp)`

- Any potential changes we might make based on these?

# Seed predation and species

Only use species specific probability, not overdispersion:

```
SP.bb.sp.prob <- mle2(taken ~ dbetabinom(prob = plogis(lprob),  
                                          theta = exp(ltheta),  
                                          size = available),  
                      parameters = list(lprob ~ species),  
                      start = list(lprob = -1,  
                                   ltheta = -1),  
                      data = SeedPred)
```

Run this and compare it with SP.bb.sp, which is better?

# Take-aways

- Model fitting is part process, part art
- This serves as an example of a few simple questions
- As usual, can start by thinking of distributions, then functions, then fits

## **Before next class:**

- Read 9.1 - 9.2 (General linear models!)
- Turn in Assignment 3 (Due Friday by midnight)